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Bayesian Mixture Cure Model for Alzheimer's Disease Progression

Eric Jiang

Institute for Applied Computational Science, Harvard University, Cambridge, Massachusetts, U.S.A

email: ericjiang@g.harvard.edu

and

Isabella Pei

Department of Computer Science and Cognitive Science, University of California, Berkeley, California, U.S.A.

email: izzypei@berkeley.edu

and

Aaron Scheffler

Department of Biostatistics and Epidemiology, UCSF, San Francisco, California, U.S.A.

email: aaronscheffler@ucsf.edu

Summary: This is the abstract written here.
Key words: A key word; But another key word; Still another key word; Yet another key word.

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1. Introduction

Alzheimer's Disease (AD) is a leading cause of mortality and disability that is marked by the accumulation of Amyloid plaques in the brain and the intracellular accumulation of tau protein tangles. Human brains can undergo pathological processes that begin years before any sign of cognitive impairment, motivating the methodology to accurately model these processes for prediction and prevention.

In order to quantifiably track these processes, clinical studies of AD use PET imaging to measure amyloid and tau via brain scans. The imaging data is collected per subject and is longitudinal over short periods of time. Each subject's data exists as a series of data points measuring amyloid burden and indexed by calendar age.

Standard Disease Progression and Survival models operate on the assumption that every individual will experience the event of accumulation given enough time. However, there is strong evidence that a proportion of individuals never develop amyloid plaque buildup throughout their lifetime.

2. Methods

2.1 *Bayesian Hierarchical Mixture Cure Model*

We now formally introduce our proposed Bayesian Mixture Cure Disease Progression Model, a statistical survival model that assumes the examined population falls under two discrete cases: amyloid accumulators and non-accumulators (individuals who do not accumulate throughout their lifetime).

2.2 *Two Latent Classes*

Consider n subjects indexed by $i = 1, \ldots, n$, with m_i longitudinal amyloid PET measurements per subject, indexed by $j = 1, \ldots, m_i$. We introduce a latent indicator Z_i distinguishing

amyloid accumulators $Z_i = 1$ from nonaccumulators $Z_i = 0$, with

$$Z_i \sim \text{Bernoulli}(\pi_i) \tag{1}$$

$$\text{logit}(\pi_i) = w_i^T \phi \tag{2}$$

where w_i are the covariates defined as: $w_i = (1, APOE4_i, Female_i)^T$ and ϕ are covariate effects. The hierarchical model places priors on the covariate effects as follows, $\phi_1 \sim \mathcal{N}(0.95, 1.0^2)$ and $\phi_i \sim \mathcal{N}(0, 0.5^2)$, for $i \in \{2, 3\}$. The intercept prior is centered at a susceptibility probability of approximately $\text{logit}^{-1}(0.95) = 0.72$ for the mean-covariate profile, while allowing substantial uncertainty in the accumulator fraction.

2.3 *Non-Accumulator Branch*

Let Y_{ij} denote the observed amyloid PET measurement for subject i at calendar age a_{ij}. We distinguish the latent amyloid trajectories of accumulators and nonaccumulators. For nonaccumulators ($Z_i = 0$), the latent amyloid level is constant at a subject-specific value b_i. For accumulators ($Z_i = 1$), the latent trajectory is represented by an age-dependent function $A_i(a)$, defined in the following section.

Conditional on class membership and the corresponding latent trajectory, we model the observed PET measurements as

$$Y_{ij} \mid Z_i = k, \ldots \sim t_4\left(\mu_{ijk}, \omega_A(\mu_{ijk})\right), \qquad k \in \{0, 1\},$$

where

$$\mu_{ij0} = b_i, \qquad \mu_{ij1} = A_i(a_{ij}).$$

Here, $\omega_A(y) > 0$ is an amyloid-dependent measurement-error scale function, specified in section 2.6. The Student-t observation model accommodates occasional large deviations from the latent trajectories.

For nonaccumulators, we assume a constant latent trajectory $A_i(a) = b_i$, where b_i follows a normal distribution with mean μ_b and variance σ_b^2, truncated above at amyloid positivity

threshold y^*. We assign $\sigma_b \sim \mathcal{N}^+(0, 0.15^2)$, further truncated to (0,0.5), where $\mathcal{N}^+$ denotes the half-normal distribution. This parameter governs between-subject variability in the latent amyloid levels of nonaccumulators, with the upper bound serving as a regularization constraint. Thus, the nonaccumulator component describes subjects whose latent amyloid burden remains below the threshold throughout the modeled age range.

2.4 *Accumulator Branch*

For accumulators, we introduce a subject-level instantaneous disease rate dependent on the current level of amyloid accumulation. Accumulator trajectories are defined by three main ideas: the time in which they start, defined as α_i, how fast they progress, defined as e^{δ_i}, and finally a shared population level rate curve, $r_A(y)$. Naturally, it makes no sense to define things like speed and onset time for non accumulators, so these covariate effects are learned conditioned on accumulator status 1.

Rate vs Value Curve

Let $r_A(A_i(a))$ be a population level rate function dependent on current amyloid level accumulation. To flexibly model the rate while enforcing positivity, we define:

$$log(r_A(y)) = B_A(y)^T\theta \tag{3}$$

where $B_A(y)$ is a a cubic B-spline basis comprising ten functions evaluated at amyloid level y, and θ are the spline coefficients. This curve is regularized with a random walk prior on its coefficients, constructed sequentially:

$$\theta_1 = -6.9,\ \theta_2 \sim \mathcal{N}(-6.9, 2^2)$$

$$\theta_i|\theta_{i-1} \sim \mathcal{N}(\theta_{i-1}, \zeta_i^2), \qquad i = 3, ..., 10$$

$$\zeta_i = \sqrt{\frac{\lambda_i}{\zeta_0}}$$

$$\lambda_i \sim Exponential(1) * \mathbb{1}\{0.01 \leqslant \lambda_i \leqslant 100\}$$

$$\zeta_0 \sim Gamma(1,1) * \mathbb{1}\{0.01 \leqslant \zeta_0 \leqslant 100\}$$

Since we learn the increment variance, $\frac{\lambda_i}{\zeta_0} = Var(\theta_i - \theta_{i-1})$, through the posterior, this prior choice encourages neighboring spline coefficients to stay next to each other while also allowing larger local differences when the data suggests it, similar to an adaptive smoothing effect. More specifically, ζ_0 is a global smoothing effect applied to the whole curve, and λ_i allows the curve to locally vary.

Why the Indicators Matter:

The truncations bound the random-walk increment variances away from zero and infinity. Conditional on λ_i and ζ_0, the increment $B_i = \theta_i - \theta_{i-1}$ follows $N(0, \lambda_i/\zeta_0)$, whose density at zero satisfies

$$f_{B_i|\lambda_i,\zeta_0}(0) = \sqrt{\frac{\zeta_0}{2\pi\lambda_i}} \longrightarrow \infty \qquad \text{as } \lambda_i \downarrow 0 \quad (\zeta_0 > 0 \text{ fixed}).$$

The exponential prior on λ_i does not suppress this divergence because its density approaches a positive constant at zero. Restricting both λ_i and ζ_0 to $[0.01, 100]$ confines the increment variance to $[10^{-4}, 10^4]$, excluding arbitrarily concentrated or diffuse conditional distributions while retaining substantial local adaptivity.

Covariate Effects on Speed

Naturally another point of interest we explore is the speed at which different subjects accumulate. Let δ_i be subject-specific log speed (the subject's accumulation growth rate is scaled by e^{δ_i}). We model this hierarchically with a linear mixed effects regression:

$$\delta_i = x_i^T \psi + \sigma_\delta z_i \tag{4}$$

where $z_i \sim N(0,1)$, $x_i = (APOE4_i, Female_i)^T$, and ψ is a vector of covariate effects on speed. Here, $\psi_i \sim \mathcal{N}(0, 0.25^2)$, for $i = 1, 2$ and $\sigma_\delta \sim \mathcal{N}^+(0,1)\mathbb{1}\{0 < \sigma_\delta < 2\}$. We model the global between subject variability parameter σ_δ as a half-normal distribution truncated at 2. Since subject-specific speed manifests multiplicative as e^{δ_i}, $\sigma_\delta = 2$ represents a speed factor multiplier of $e^2 = 7.389$. The upper truncation therefore serves as a weak guardrail against

implausibly large residual speed heterogeneity.

Covariate Effects on Accumulation Launch Time, α_i

We define our subject level accumulation launch time, α_i, as a Weibull accelerated failure time (AFT) model:

$$\alpha_i = \alpha_{min} + e^{w_i^T \gamma} * E_i^{\sigma_\alpha}, \ \alpha_i \leqslant \alpha_{max} \tag{5}$$

where $E_i \sim Exponential(1)$, $\alpha_{min} = 25$, and $\alpha_{max} = 125$. $w_i = (1, APOE4_i, Female_i)^T$ is the same covariate vector with an intercept term defined from before. Here, γ denotes the regression coefficients governing departure age, while σ_α controls between-subject variation in departure timing. We assign $\gamma_1 \sim \mathcal{N}(\log 45, 0.75^2)$, centering the baseline Weibull scale age at $25 + \exp(\log 45) = 70$ years. For the remaining coefficients, we specify $\gamma_k \sim \mathcal{N}(0, 0.25^2)$, (k=2,3), corresponding to a one-prior-standard-deviation multiplicative effect of $\exp(0.25) \approx 1.28$ on the Weibull time scale. We assign $\sigma_\alpha \sim$ Gamma$(2, 8)$, truncated to $(0, 2)$, using the shape-rate parameterization. The upper truncation restricts extreme heterogeneity.

2.5 *Initial condition motivation & solution of ODE*

We model the latent amyloid trajectory by a first order separable ordinary differential equation(ODE):

$$\frac{dA_i}{da} = e^{\delta_i} r_A(A_i(a)) \tag{6}$$

Naturally, an initial condition is required to specify a unique solution to the ODE. Consider the general case where each subject has their own unknown launch level, $A_i(\alpha_i) = c_i$. Defining the strictly increasing disease-time transformation $G_A(y) = \int_{y^*}^{y} r_A(u)^{-1}\, du$, integration of the separable ODE yields:

$$G_A(A_i(a)) = e^{\delta_i} a + \left[G_A(c_i) - e^{\delta_i}\alpha_i\right], \qquad a \geqslant \alpha_i.$$

A more rigorous derivation of this is provided in Section 7 for clarity. When observations are restricted to the post-departure portion of a trajectory, the launch level and departure

age enter through a single combination, (α_i, c_i), through $G_A(c_i) - e^{\delta_i}\alpha_i$, making them non-identifiable in that region. Thus we fix $A_i(\alpha_i) = \mu_b$ and all subjects are held at a fixed launch/floor level μ_b instead of having a subject-specific intercept. Note that $G_A(y)$ is strictly monotone, thus its inverse exists on the entire image space. Inverting the disease-time transformation yields the following representation of the subject-specific trajectory:

$$A_i(a) = \begin{cases} \mu_b, & a < \alpha_i \\ G_A^{-1}[G_A(\mu_b) + e^{\delta_i}(a - \alpha_i)], & a \geqslant \alpha_i, \end{cases} \tag{7}$$

$$\mu_b \sim \mathcal{N}(0.45, 0.15^2)\mathbb{1}\{Y_L < \mu_b < y^*\}$$

$Y_L = 0.30$ SUVR denotes the lower boundary of the numerical amyloid-level grid. Truncating μ_b to (Y_L, y^*) ensures that the common accumulation launch level lies within the numerical domain and below the amyloid positivity threshold.

Furthermore, $G_A(y)$ can be interpreted as the signed number of disease-years needed to travel from positivity threshold y^* to current level y at unit speed. Note that the clock's construction let's us interpret the function as:

For amyloid positive y: $G_A(y) > 0$. For amyloid negative y: $G_A(y) < 0$, and $G_A(y*) = 0$. This way $-G_A(\mu_b)$ is the baseline transit budget, interpreted as the time it takes for an accumulator to travel from the common floor μ_b to the positivity threshold y^* at unit specific speed. We can then derive a subject specific positivity age a_i^*, by the fact that $A_i(a_i^*) = y^*$

$$a_i^* = \alpha_i - G_A(\mu_b)e^{-\delta_i} \tag{8}$$

2.6 *Error Curve* $\omega_A(y)$

The (Measurement Error) Scale function We similarly construct our measurement scale function with a cubic B-spline with 8 basis functions on the log scale,

$$\log \omega_A(y) = B^{\nu}(y)^T\nu, \qquad \nu = [\nu_1, ..., \nu_8]^T \tag{9}$$

but instead of an adaptive smoothing random walk prior on the coefficients, we use a fixed width random walk constructed sequentially as follows:

$$\nu_1 \sim \mathcal{N}(-3.4, 1.0^2)\mathbb{1}\{-9 < \nu_1 < 1\}$$

$$\nu_j \sim \mathcal{N}(\nu_{j-1}, 0.5^2)\mathbb{1}\{-9 < \nu_j < 1\}, \qquad j = 2, ..., 8$$

The truncation is simply acting as a numerical guardrail against extremely small or large measurement error scales. We model measurement error as a function of amyloid level because residual distributions are generally hetero-skedastic and depend on the current signal of amyloid (1) Measurement error has a fixed width random walk because the measurement scale is inherently a nicely behaved, slow varying function that does not need the extra flexibility that an adaptive smoothing prior provides.

2.7 *Constructing the Likelihoods*

We define the likelihood function,

$$L_i = \pi_i L_{i1} + (1 - \pi_i) L_{i0} \tag{10}$$

where $L_{ik} = P(Y_i | Z_i = k)$.

3. Posterior inference through Markov chain Monte Carlo

3.1 ***Sampler Implementation***

Posterior inference was implemented in R using the NIMBLE framework, which permits user-defined probability distributions and customized MCMC sampling schemes. The hierarchical model was specified using nimbleCode, constructed with nimbleModel, and compiled with compileNimble prior to MCMC execution. Numerical components required by the disease-time transformation, including trapezoidal integration, interpolation, inversion of the monotone disease clock, and evaluation of the Student-t log density, were implemented as compiled

nimbleFunctions. The disease-time transformation is evaluated on a uniform amyloid-level grid spanning 0.30 to 1.70 SUVR, with spacing 0.002. Numerical integration is performed using the cumulative trapezoidal rule, and the strictly increasing disease clock is inverted using monotone linear interpolation.

Two user-defined distributions were used to represent components that are not conveniently expressed through standard BUGS syntax. First, rather than sampling the subject-specific departure age α_i directly, we employ a reparameterization based on the latent trajectory-location variable $\widetilde{x}_i$ at the subject's mean observation age. Departure age is recovered deterministically through the disease-time transformation, while the user-defined distribution 'dxtilde' specifies the density of $\widetilde{x}_i$ induced by the Weibull AFT prior on α_i through a change of variables. No additional prior is assigned to $\widetilde{x}_i$. The transformation and corresponding Jacobian are detailed in Web Appendix A. Second, dsubjMix evaluates the subject-level longitudinal likelihood by propagating the accumulator trajectory through the disease clock and combining the accumulator and non-accumulator likelihood contributions using a numerically stable log-sum-exp calculation. The latent susceptibility indicator Z_i is therefore marginalized rather than sampled explicitly.

We modify NIMBLE's default MCMC configuration for parameters with substantial posterior dependence or constrained support. The measurement-error spline coefficients ν, the non-intercept departure-age coefficients $\gamma_{2:3}$, and the speed coefficients ψ are sampled in separate adaptive-factor slice `AF_slice` blocks. The departure-age intercept γ_1 is jointly sampled with the free rate-curve coefficients $\theta_{2:K}$, reflecting their posterior dependence through the disease-time scale. The scalar scale and floor parameters σ_δ, σ_α, μ_b, and σ_b are updated using univariate slice samplers. At the subject level, the non-accumulator level b_i is updated using a univariate slice sampler. The latent trajectory-location variable $\tilde{x}_i$ and subject-specific speed random effect z_i are instead updated jointly using a random-walk block

sampler, because these quantities exhibit strong posterior dependence: later departure can be compensated by faster progression, and conversely. Joint updating improves movement along this posterior ridge relative to separate component-wise proposals. Parameter states that violate the trajectory domain or departure-age constraints are assigned log-density $-\infty$. This prevents the sampler from exploring numerically infeasible regions of the parameter space. Initial values are constructed to satisfy these constraints, and subsequent chains are initialized from feasible perturbations of the baseline initialization so that convergence can be assessed from distinct starting points.

3.2 ***Posterior Summaries and Diagnostics***

3.2.1 *Posterior Inference.* Posterior inference is conducted on global and subject specific parameters with 95% credible intervals and recorded mean and median posterior values.

3.2.2 *Convergence Metrics.* MCMC convergence is assessed using rank-normalized split-$\hat{R}$ and bulk and tail effective sample sizes (ESS). Splitting each chain increases sensitivity to within-chain nonstationarity, while rank normalization improves robustness to non-Gaussian posterior marginals. Bulk ESS evaluates sampling efficiency for central posterior summaries, whereas tail ESS assesses the reliability of posterior tail quantiles. We use $\hat{R} < 1.01$ as a diagnostic threshold and examine effective sample sizes to identify parameters exhibiting substantial autocorrelation or inefficient mixing.

4. Prediction Methodology

Suppose we observe a new subject with covariates x_* and longitudinal scans $\mathcal{D}_* = \{(a_{*j}, Y_{*j})\}_{j=1}^{m_*}$. It's often of interest to consider posterior inference for accumulator status, accumulation departure age, and progression speed among other clinically relevant things.

Let η_* be a vector of the new subjects latent variables of clinical interest. Let $\mathcal{D}_{train}$ be the

training data and Θ be a vector of our global parameters grouped together. Then $P(\Theta|\mathcal{D}_{train})$ is our fitted posterior. Then after observing the new subjects PET scans, the subject specific posterior predictive distribution is:

$$p(\eta_*|\mathcal{D}_*, \mathcal{D}_{train}) = \int p(\eta_*|\mathcal{D}_*, \Theta)p(\Theta|\mathcal{D}_*, \mathcal{D}_{train})d\Theta \tag{11}$$

which is not available in closed form but can be sampled using MCMC.

5. Simulation Studies

We utilize a comprehensive set of simulations to evaluate the functionalities/behavior of our proposed Mixture-cure model under settings similar to our motivating ADNI data.

5.1 *Data Generation*

In our simulations, we

number of subjects: 1101 population parameters –¿ fitted posterior medians covariates x1 and x2: simulated independently under distributions from a bernoulli(0.407) and bernoulli(0.489) distribution [MIGHT CHANGE TO bernoulli(0.5)] and centered them (x1 - 0.427, x2 - 0.503)

baseline age simulated from normal(72.5, 7.4) distribution, truncated [51, 94]

number of observations is generated from a discrete distribution 2, 3, 4, 5, 6, 7 with probabilities (487, 296, 155, 83, 64, 16) / 1101 as observed in our real data. gaps between scans are simulated from normal(2.05, 0.15) distribution based on observed mean of 3 years –¿ visit ages = baseline age + cumulative sum of gaps

rate curve:

for each subject i, susceptibility Z simulated from bernoulli(pi) distribution where pi = plogis(Xg x phi). alpha

for non accumulators, b drawn from normal(mu b, sigma b) distribution truncated above at threshold

for accumulators: use eq

5.2 *Evaluation Metrics*

5.3 *Implementation Details*

6. Results

7. Derivations and Proofs

Start with the previously defined ODE:

$$\frac{dA_i}{da} = e^{\delta_i} r_A(A_i(a))$$

separating and integrating both sides gives,

$$\int_{c_i}^{A_i(a)} \frac{du}{r_A(u)} = \int_{\alpha_i}^{a} e^{\delta_i} dt$$

thus we define $G_A(y) = \int_{y*}^{y} \frac{du}{r_A(u)}$, allowing us to use Fundamental Theorem of Calculus:

$$G_A(A_i(a)) - G_A(c_i) = (a - \alpha_i)e^{\delta_i}$$

giving equation,

$$G_A(A_i(a)) = G_A(c_i) + (a - \alpha_i)e^{\delta_i} = ae^{\delta_i} + [G_A(c_i) - \alpha_i e^{\delta_i}]$$

Note that $G_A(y)$ is a strictly monotone function, thus its inverse exists on the entire image space. Thus inverting both sides gives us an exact closed form:

$$A_i(a) = G_A^{-1}[G_A(c_i) + e^{\delta_i}(a - \alpha_i)]$$

8. Discussion

Put your final comments here.

Acknowledgements

Supplementary Materials

Web Appendix A, referenced in Section 2, is available with this paper at the Biometrics website on Wiley Online Library.

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Appendix

Title of appendix

Put your short appendix here. Remember, longer appendices are possible when presented as Supplementary Web Material. Please review and follow the journal policy for this material, available under Instructions for Authors at `http://www.biometrics.tibs.org`.